

PocketSmart AI

Your Smart Budget & Recommendation Assistant | Technical Documentation

1. PROJECT OVERVIEW

Managing budgets across different life needs—like home decor, event planning, or jewelry shopping—can be overwhelming due to the wide variety of products, platforms, and price ranges. **PocketSmart AI** addresses this challenge through a GenAI-powered, cross-platform recommendation system that delivers personalized, budget-based suggestions for products and services.

Using **Gemini 1.5 Flash Pro**, PocketSmart AI analyzes user preferences, budgets, and contextual needs to generate curated recommendations from popular platforms such as Amazon, Flipkart, IKEA, Swiggy, Zomato, OYO, and more. The platform helps users plan interior designs, parties, and even select matching jewelry for special occasions—all within their defined budgets.

Key Use-Case Scenarios

Scenario	Description & Workflow
1. Home Interior Planning	Users select the Home Budget Planner, enter budget, room types (Living Room, Kitchen, Bedroom), and quantities for lights, fans, dining tables, etc. Gemini 1.5 Flash Pro recommends cost-effective options from platforms like IKEA and Amazon balanced across style, functionality, and cost.
2. AI Party Budget Planning	Users input total budget, guest count, event type, and venue details. The AI proportionally allocates budget across catering, decoration, and entertainment using vendors from Swiggy, Zomato, and OYO.
3. Jewelry Recommendations	Allows users to enter budget, occasion, and style preferences with an optional outfit image upload. Gemini analyzes outfit color, design, and aesthetics to provide complementary jewelry suggestions from Amazon and Flipkart.

2. SYSTEM ARCHITECTURE & CORE TECHNOLOGIES

PocketSmart AI is built as an intelligent, modular platform combining a lightweight backend with advanced generative AI capabilities.

Core Tech Stack:

- **FastAPI / Flask:** Backend service layer for handling API routes, sessions, and request orchestration.
- **Gemini 1.5 Flash Pro:** Google's multimodal AI model for budget reasoning, multimodal (text + image) analysis, and structured output generation.
- **Frontend (HTML/CSS + JS):** Interactive user interface built with Jinja2 templates for seamless input capture and response rendering.

- **E-Commerce Integrations:** Dynamic search links generation for platforms like Amazon, Flipkart, IKEA, Swiggy, Zomato, BookMyShow, etc.

Component Architecture Summary

Component	Responsibilities
Frontend UI	Web interface with dynamic forms for inputs (budgets, counts, image uploads) and display of recommendations.
Backend Service	Manages session tokens, user authentication, routing (/generate-home, /generate-party, /generate-jewelry), and history logs.
Gemini AI Layer	Processes prompts, evaluates constraints, executes multimodal image analysis, and returns structured JSON responses.

3. PROJECT WORKFLOW & DEVELOPMENT MILESTONES

Milestone 1: Gemini AI Initialization

Configured Google Cloud access, enabled Gemini API keys, and verified multimodal functionality using Python SDK. Ensured text-only and image+text prompts function with high accuracy.

```
# Sample Initialization snippet
import google.generativeai as genai

genai.configure(api_key=os.getenv("GOOGLE_API_KEY"))
model = genai.GenerativeModel('gemini-1.5-flash')
```

Milestone 2: Core Functionality & Prompt Engineering

Developed modular recommendation functions in gemini_utils.py handling budget allocation, currency calculations (INR/USD), JSON output extraction, and fallback mechanisms when API data is limited.

Milestone 3: FastAPI Backend Implementation

Implemented secure authentication (JWT tokens, password hashing), CORS middleware, static file serving for outfit uploads, and structured API routes:

- POST /home-budget - Generates detailed furniture and lighting budget distribution.
- POST /party-budget - Computes guest-wise cost breakdown for food, venue, and decor.
- POST /jewelry-budget - Evaluates dress imagery for color matching and recommendations.
- GET /recommendation-history - Tracks past user queries and saved budget plans.

Milestone 4: UI Development & Frontend Pages

Created intuitive web pages using Jinja2 templates:

- **Dashboard Page:** Overview of active sessions and recent budget searches.

- **Planner Interfaces:** Interactive forms with dynamic field additions.
- **Results Display:** Card-based recommendations with direct shopping redirect buttons.

Milestone 5: Testing, Optimization & Deployment

Executed end-to-end testing across varied budget limits and real-world inputs. Fine-tuned prompts to enforce strict budget adherence and fallback suggestions.

4. CONCLUSION

PocketSmart AI effectively combines Generative AI with web frameworks to make financial and shopping decisions effortless. By delivering accurate, budget-conscious, multi-platform recommendations, it empowers users to achieve their interior design, party planning, and personal styling goals without financial uncertainty.

5. PROJECT VERIFICATION & SIGN-OFF

Module / Feature	Status	Verification Note
Gemini 1.5 API Integration	Completed	Multimodal (Text + Outfit Image) validated
FastAPI / Flask Backend Routing	Completed	Endpoints for Home, Party, Jewelry active
UI & Recommendation Display	Completed	Responsive layout with dynamic shopping links
User Session & History Management	Completed	JWT Auth & persistent history tracking enabled